

Open-system model for light-shift-driven multiparameter spin metrology

1 Setup

We consider a single spin J (one atom of a product ensemble of N_a atoms; for $J = 3$ this is the Cs $F = 3$ hyperfine manifold) evolving under a closed-system Hamiltonian that is the sum of a static field to be estimated, an rf control rotation with a piecewise-constant random phase, and a nonlinear term generated by the tensor light shift of an off-resonant, linearly polarized beam. With $\hbar = 1$ and time in units of $1/\Omega$,

$$H(t) = \underbrace{\mathbf{w} \cdot \mathbf{J}}_{\text{field to estimate}} + \underbrace{\Omega [\cos \phi(t) J_x + \sin \phi(t) J_y]}_{\text{rf control}} + \underbrace{\frac{\beta}{2J} J_z^2}_{\text{tensor light shift}}, \quad (1)$$

where $\mathbf{w} = (w_x, w_y, w_z)$ is the three-component parameter vector and $\phi(t)$ is constant on blocks of `random_phase_steps` time steps and drawn uniformly on $[0, 2\pi)$ for every realization. The code's parameter `beta` enters as $\beta/2J$, as written.

2 Light-induced decoherence

The same beam that produces the J_z^2 term scatters photons. In the far-detuned limit (detuning large compared with the excited-state hyperfine splitting Δ'_{HF} but small compared with the fine-structure splitting) the scattering amplitude acts on the electron spin alone. For linear polarization the no-spin-flip (Rayleigh) part is a scalar and does not decohere the ground manifold. The spin-flip (Raman) part, projected onto the F manifold, is proportional to $J_{\pm}$ by the Wigner–Eckart theorem. Neglecting Raman loss to the other hyperfine manifold and any decoherence from a separate readout beam, the single decoherence channel is therefore the pair of jump operators

$$L_+ = \sqrt{\gamma} J_+, \quad L_- = \sqrt{\gamma} J_-, \quad J_{\pm} = J_x \pm iJ_y. \quad (2)$$

Both the tensor light shift and the scattering rate scale with intensity over detuning squared, $\beta \propto I \Delta'_{\text{HF}}/\Delta^2$ and $\gamma_s \propto I \Gamma/\Delta^2$, so their ratio is a property of the atom and the transition, not of the beam. We parametrize this by a fixed figure of merit

$$\eta \equiv \frac{\beta}{\gamma}, \quad \gamma = \frac{\beta}{\eta}, \quad \eta \sim c \frac{\Delta'_{\text{HF}}}{\Gamma} = \mathcal{O}(10\text{--}100), \quad (3)$$

with c an order-one Clebsch–Gordan factor obtainable from the ground-state polarizability and optical-pumping formalism of Deutsch and Jessen (Opt. Commun. **283**, 681 (2010)). Consequently the nonlinearity cannot be increased without increasing the decoherence in proportion, and a sweep over β is simultaneously a sweep over γ .

3 Master equation (Schrödinger picture)

The Lindblad master equation for the single-atom density matrix is

$$\frac{d\rho}{dt} = -i[H(t), \rho] + \gamma \left(J_+ \rho J_- - \frac{1}{2} \{J_- J_+, \rho\} \right) + \gamma \left(J_- \rho J_+ - \frac{1}{2} \{J_+ J_-, \rho\} \right). \quad (4)$$

Writing $J_{\pm} = J_x \pm iJ_y$, the cross terms cancel and the dissipator collapses to

$$\mathcal{D}[\rho] = 2\gamma \sum_{k=x,y} \left(J_k \rho J_k - \frac{1}{2} \{J_k^2, \rho\} \right) = -\gamma \left([J_x, [J_x, \rho]] + [J_y, [J_y, \rho]] \right), \quad (5)$$

a double-commutator (dephasing-type) channel along x and y , each at rate 2γ . It commutes with rotations about z , conserves the trace, and is unital: the maximally mixed state $\mathbb{K}/(2J+1)$ is its fixed point. Here $J_- J_+ + J_+ J_- = 2(J_x^2 + J_y^2) = 2[J(J+1) - J_z^2]$.

4 Master equation (Heisenberg picture)

Expectation values can equally be computed by propagating the observable with the adjoint generator,

$$\langle O \rangle(t) = \text{Tr}[O e^{t\mathcal{L}}(\rho_0)] = \text{Tr}[e^{t\mathcal{L}^\dagger}(O) \rho_0], \quad (6)$$

with

$$\begin{aligned} \frac{dO}{dt} &= i[H(t), O] + \gamma \left(J_- O J_+ - \frac{1}{2} \{J_- J_+, O\} \right) + \gamma \left(J_+ O J_- - \frac{1}{2} \{J_+ J_-, O\} \right) \\ &= i[H(t), O] - \gamma \left([J_x, [J_x, O]] + [J_y, [J_y, O]] \right). \end{aligned} \quad (7)$$

The dissipator takes the same form in both pictures because the jump set $\{J_+, J_-\}$ is closed under Hermitian conjugation. This is the form integrated in the simulation: the observable $O = J_y$ (or J_z) is propagated and the signal is $\text{Tr}[O(t)\rho_0]$ with ρ_0 the initial spin coherent state.

Exact relaxation of the first moments

Because $\mathcal{D}^\dagger$ maps the vector operator $\mathbf{J}$ onto itself,

$$\mathcal{D}^\dagger[J_x] = -\gamma J_x, \quad \mathcal{D}^\dagger[J_y] = -\gamma J_y, \quad \mathcal{D}^\dagger[J_z] = -2\gamma J_z, \quad (8)$$

so for any J the Bloch vector obeys

$$\frac{d}{dt} \langle J_x \rangle = (\text{coherent}) - \gamma \langle J_x \rangle, \quad (9)$$

$$\frac{d}{dt} \langle J_y \rangle = (\text{coherent}) - \gamma \langle J_y \rangle, \quad (10)$$

$$\frac{d}{dt} \langle J_z \rangle = (\text{coherent}) - 2\gamma \langle J_z \rangle. \quad (11)$$

The coherent part is not closed at first order because the J_z^2 term couples $\langle \mathbf{J} \rangle$ to second moments; it is handled numerically below. The parameter γ is therefore the transverse relaxation rate, and 2γ the longitudinal one.

5 Measurement model

At each time step n the recorded datum is

$$M_n = \langle O \rangle_n + \xi_n, \quad \xi_n \sim \mathcal{N}(0, \sigma^2), \quad \sigma = \frac{1}{\sqrt{N_p} G N_a}, \quad (12)$$

i.e. the ensemble-averaged single-atom expectation value with Gaussian photon shot noise from a separate readout beam of N_p photons per step, Faraday gain G , and N_a atoms. Back-action of the readout beam is not included. The estimator is the least-squares fit of $\mathbf{w}$ to $\{M_n\}$ using the same open-system forward model, with γ treated as known (it is fixed by the chosen β), and its covariance is compared with the classical Cramér–Rao bound $\text{Cov}(\hat{\mathbf{w}}) \geq F^{-1}$,

$$F_{ab} = \frac{1}{\sigma^2} \sum_n \frac{\partial \langle O \rangle_n}{\partial w_a} \frac{\partial \langle O \rangle_n}{\partial w_b}. \quad (13)$$

6 Numerical integration

Time is discretized with step δt . With $U_w = e^{-i\delta t \mathbf{w} \cdot \mathbf{J}}$ and $U_c[n] = e^{-i\delta t H_c[n]}$ the control step for block n , define $U_n = U_c[n] U_w$. The Hamiltonian and dissipative parts are split to first order,

$$\text{Schrödinger:} \quad \rho_{n+1} = e^{\delta t \mathcal{D}} (U_n \rho_n U_n^\dagger), \quad (14)$$

$$\text{Heisenberg:} \quad O_{n+1} = U_n^\dagger e^{\delta t \mathcal{D}^\dagger} (O_n) U_n. \quad (15)$$

The map $A \equiv e^{\delta t \mathcal{D}^\dagger}$ is built once per realization as a $(2J+1)^2 \times (2J+1)^2$ matrix acting on the vectorized observable (the exact solution of the dissipative part over one step), so no approximation beyond the first-order splitting is made. That splitting is of the same order as the pre-existing split between U_w and U_c . Because A is linear and independent of $\mathbf{w}$, the Fréchet-derivative recursion for $\partial O_n / \partial w_a$, needed for the Fisher matrix, is unchanged apart from passing O_n and $\partial O_n / \partial w_a$ through A first. Setting $\gamma = 0$ (no `--eta`) recovers the closed-system code exactly.

Expected consequence. The signal derivatives $\partial \langle O \rangle_n / \partial w_a$ decay on the time scale $1/\gamma$, so the accumulated Fisher information saturates near $T \sim 1/\gamma$ instead of growing polynomially. At a fixed total time T the useful nonlinearity is bounded by $\beta \lesssim \eta/T$, and the question becomes whether the nonlinearity improves multiparameter estimation enough to pay for the decoherence it necessarily brings.

7 Assumptions and how to relax them

The model above rests on two simplifications, adopted deliberately for the first study. Both can be removed within the same numerical scheme.

A1: the readout beam does not decohere the atoms

Relaxed in Section 8 (script v6).

The measurement model treats the N_p readout photons per step as pure shot noise. A readout beam scatters photons too, through the same Raman channel (2), so it adds to γ a rate γ_r set by the readout intensity rather than by β . Because Faraday signal and scattering both scale as

$1/\Delta^2$, the ratio of measurement signal-to-noise to scattering is independent of detuning and fixed by the resonant optical depth $\text{OD} = N_a \sigma_0 / A$: up to order-one polarization and branching factors, the number of photons scattered per atom over the whole run is

$$P_{\text{sc}} \simeq \frac{\text{SNR}_{\text{tot}}^2}{N_a \text{OD}}, \quad \text{SNR}_{\text{tot}}^2 = G^2 N_a^2 N_p^{\text{tot}} = \sum_n \sigma^{-2} \quad (16)$$

per unit of $\langle J \rangle$. With the values used in the simulations ($G = 8.9 \times 10^{-7}$, $N_a = 2 \times 10^5$, $N_p^{\text{tot}} = 9.6 \times 10^8$) this gives $\text{SNR}_{\text{tot}}^2 \approx 3 \times 10^7$ and hence $P_{\text{sc}} \sim 10^2 / \text{OD}$, i.e. $\gamma_r T \sim 10^2 / \text{OD}$. For comparison the light-shift beam contributes $\gamma T = \beta T / \eta$, which for $T = 14 / \Omega$ and $\eta \sim 30$ is $\approx 0.5 \beta$. So unless $\text{OD} \gtrsim 10^3$ (a cavity, or a very dense cloud), the readout beam decoheres the atoms at least as strongly as the light-shift beam does for $\beta \lesssim 1$, and assumption A1 is not self-consistent with the assumed shot-noise level. Relaxing it costs one extra parameter, OD, and one change to the model: replace $\gamma \rightarrow \gamma + \gamma_r$ in (2) with $\gamma_r = P_{\text{sc}} / T$, independent of β . This also introduces the usual continuous-measurement trade-off between N_p and γ_r that the present model does not see. (In the single-beam scheme of Deutsch and Jessen the light-shift and readout beams are the same beam, so β , N_p and γ are all tied to one intensity.)

A2: no loss from the F manifold

Relaxed in Section 9 (script v7).

The Raman process that gives (2) also transfers atoms to the other ground hyperfine manifold ($F = 4$ for Cs $F = 3$) with a branching ratio of order one, so loss occurs at a rate $\gamma_L \sim \gamma$. Within the F -manifold Hilbert space this is a non-trace-preserving term

$$\frac{d\rho}{dt} \supset -\frac{1}{2} \{K_L, \rho\}, \quad K_L = \sum_q L_q^{L\dagger} L_q^L = a + c J_z^2, \quad (17)$$

where $L_q^L \propto P_{F'} s_q P_F$ are the jump operators into F' and the form of K_L follows from z -symmetry and the $m \rightarrow -m$ reflection symmetry of π -polarized light. If the lost atoms are dark to the readout, the unnormalized $\text{Tr}[O\rho]$ is automatically the measured signal, and in the simplest uniform approximation $K_L = \gamma_L \mathbb{K}$ this is just a known factor $e^{-\gamma_L t}$ multiplying the signal and its derivatives. In the code this is one additional w -independent factor per step, so the fit and Fisher recursion are again unchanged. The complication that this approximation hides is that atoms in F' are not dark: they Faraday-rotate the readout with a different (roughly opposite) sign and evolve under a different light shift. Treating that properly requires enlarging the state space to $F \oplus F'$ (dimension 16 for Cs), which the superoperator scheme of Section 6 supports at the cost of a 256×256 dissipative map.

8 One beam: light shift, decoherence and readout from the same probe

In the actual scheme there is a single off-resonant, linearly polarized probe. It produces the tensor shift, it scatters photons, and its Faraday rotation is the readout. With photon flux Φ and detuning Δ , in the far-detuned limit,

$$\beta \propto \frac{\Phi \Delta'_{\text{HF}}}{\Delta^2}, \quad \gamma \propto \frac{\Phi \Gamma}{\Delta^2}, \quad G \propto \frac{1}{\Delta}, \quad \sigma^2 = \frac{1}{N_p G^2 N_a^2} \propto \frac{\Delta^2}{\Phi}, \quad (18)$$

where G is the rotation per atom per unit spin and $N_p \propto \Phi$ the photons per measurement interval δt . All four depend on Φ and Δ only through Φ/Δ^2 , so the probe strength is a single knob and two ratios are fixed by the physics:

$$\frac{\beta}{\gamma} = \eta, \quad \gamma \sigma^2 \delta t = \frac{1}{\text{OD}_{\text{eff}} N_a}, \quad \text{OD}_{\text{eff}} = c \frac{N_a \sigma_0}{A}, \quad (19)$$

with c an order-one factor from the vector polarizability and the spin-flip branching ratio. The second relation is the readout back-action of Section 7 made exact: the photons that carry the Faraday signal are the photons that scatter. Taking β as the sweep variable, the script `v6` derives per β

$$\gamma = \frac{\beta}{\eta}, \quad \sigma(\beta) = \sqrt{\frac{\eta}{\text{OD}_{\text{eff}} N_a \delta t \beta}}. \quad (20)$$

A stronger probe brings more nonlinearity, more decoherence and *less* shot noise; $\beta \rightarrow 0$ is no light at all, with no decoherence but no measurement either, so $\beta = 0$ is excluded from the sweep and there is no decoherence-free reference. The former constants G and N_p are absorbed into OD_{eff} ; with the values used in `v4/v5` and $\eta = 30$, $\text{OD}_{\text{eff}} = 10$, the old noise level $\sigma = 0.021$ corresponds to $\beta \approx 33$, a far stronger probe than any β in the `v5` sweep. The two physics inputs are now η (atom) and OD_{eff} (sample); both are placeholders in the launcher until derived and measured.

Expected shape of the optimum. The Fisher information per step is $(\partial \langle O \rangle / \partial w)^2 / \sigma^2 \propto \beta (\partial \langle O \rangle / \partial w)^2$, and the derivative decays on the scale η/β . For $T \ll \eta/\beta$ the information grows like βT^3 (more probe is better), for $T \gg \eta/\beta$ it saturates like η^3/β^2 (less probe is better), so the optimum sits at a finite $\beta_{\text{opt}} \sim \eta/T$ for every T . Unlike `v5`, where $\beta = 0$ eventually won because it was decoherence-free, the one-beam model always has a nonzero optimal probe strength.

Control experiment. Because the nonlinearity cannot be switched off physically without switching off the probe, `v6` has a flag `--tensor-scale` that multiplies the J_z^2 term in the Hamiltonian only, keeping the beam-derived γ and σ . Running the same sweep with `--tensor-scale 0` isolates what the nonlinear dynamics contributes to the estimation at fixed decoherence and fixed shot noise.

9 Loss, and the light-induced coefficients from atomic structure

The coefficients that Sections 2 and 8 left as placeholders (η , the order-one factors in OD_{eff}) follow from the atomic structure. The script `cs_light_shift_scattering_ratios.py` builds, for Cs ($I = 7/2$) on the D1 or D2 line and a π -polarized far-detuned probe, the ground-state light-shift operator and the optical-pumping jump operators by adiabatic elimination of the excited manifold (Steck matrix elements, closure and normalization checked, vector-to-scalar polarizability ratio converging to $\pm 1/4$ on D1 and $\pm 1/8$ on D2). Writing γ_s for the total photon scattering rate of the spin coherent state, the far-detuned values are

line, F	$\eta_s = \beta/\gamma_s$	r_{within}	$K_L/\gamma_s = a_L + c_L m^2$	b/a	$\eta_{\text{tot}} = \beta/R_{\text{tot}}$
D2, $F = 3$	7.4	0.0055	$0.204 + 0.0107 m^2$	0.127	33
D2, $F = 4$	-9.5	0.0052	$0.130 + 0.0104 m^2$	-0.123	-61
D1, $F = 3$	-100	0.0110	$0.412 + 0.0264 m^2$	-0.243	-217
D1, $F = 4$	122	0.0115	$0.254 + 0.0170 m^2$	0.257	408

Here $r_{\text{within}}\gamma_s$ is the rate of the within-manifold spin-flip channel (2), K_L the rate operator of Raman transfer out of the manifold, b/a the vector-to-scalar polarizability ratio per unit m that sets the Faraday signal, and $R_{\text{tot}} = r_{\text{within}}\gamma_s + \langle K_L \rangle$ the initial relaxation rate of $\langle F_x \rangle$. A negative η_s is a tensor shift of the opposite sign. Two things stand out: the relaxation is 96–98 % loss and only a few per cent spin flips, so the channel of Sections 2–5 is the minor one; and $\eta_{\text{tot}} = 33$ for D2, $F = 3$ is essentially the placeholder $\eta = 30$ used so far, while D1, $F = 4$ offers twelve times more nonlinearity per unit of relaxation.

Master equation with loss. The loss adds a trace-decreasing term with no jump operator, since the atom leaves the manifold:

$$\frac{d\rho}{dt} = -i[H, \rho] + \mathcal{D}_{\text{sf}}[\rho] - \frac{1}{2}\{K_L, \rho\}, \quad \frac{dO}{dt} = i[H, O] + \mathcal{D}_{\text{sf}}^\dagger[O] - \frac{1}{2}\{K_L, O\}, \quad K_L = \gamma_s(a_L + c_L J_z^2), \quad (21)$$

with $\mathcal{D}_{\text{sf}}$ the channel (5) at rate $r_{\text{within}}\gamma_s$. The step map $A = e^{\delta t \mathcal{D}^\dagger}$ of Section 6 absorbs both terms and stays $\mathbf{w}$ -independent; on the identity it acts as $e^{-\delta t K_L}$, so $\text{Tr}[O(t)\rho_0]$ is the survival probability times the conditional expectation, which is the collective Faraday signal if lost atoms are dark. Survivors evolve coherently: loss only scales the signal, unlike the spin-flip channel that scrambles it. The remaining approximation of the light model is that atoms transferred to F' are not in fact dark; they carry a vector response of opposite sign and a different light shift.

Parametrization (script v7). With the probe strength swept as β ,

$$\gamma_s = \frac{\beta}{|\eta_s|}, \quad \gamma_{\text{sf}} = r_{\text{within}}\gamma_s, \quad K_L = \gamma_s(a_L + c_L J_z^2), \quad \sigma^2 = \frac{1}{(b/a)^2 \text{OD}_{\text{res}} N_a \delta t \gamma_s}, \quad (22)$$

where $\text{OD}_{\text{res}} = N_a \sigma_{\text{res}}/A$ is the resonant optical depth of the probe transition, the only sample input besides N_a . In terms of Section 8, $\text{OD}_{\text{eff}} = (b/a)^2 \text{OD}_{\text{res}}/(R_{\text{tot}}/\gamma_s)$, which is 0.07 (D2, $F = 3$) to 0.22 (D1, $F = 4$) of OD_{res} . The $\text{OD}_{\text{eff}} = 10$ of the first one-beam run therefore corresponds to $\text{OD}_{\text{res}} \approx 140$; a free-space cloud of 2×10^5 atoms in a $50 \mu\text{m}$ beam has OD_{res} of order 6–9; both were run (Section 10).

Cramér–Rao bookkeeping. The simulation fits every realization on its own control sequence. The bound that such an estimator can reach, averaged over realizations, is $\mathbb{E}[F_i^{-1}]$, not $[\mathbb{E}F_i]^{-1}$ (Jensen); with strong decoherence the information comes from a short window in which the particular control phases matter, and the two differ by an order of magnitude (a factor 17 at $\beta = 20$, $\eta = 30$). Scripts v6 and v7 save both; the v7 runs confirm that the estimator reaches $\mathbb{E}[F_i^{-1}]$ to within a few per cent once its fits have converged (Section 10).

10 Results of the simulations

All runs use $J = 3$, observable J_y , $T = 14/\Omega$, 300 fits per (β, cutoff) with 40 cutoffs, and 20 independent draws of $\mathbf{w}_{\text{true}}$ (one per cluster array task); statistics are averaged over the 20 draws. Fits whose least-squares cost lies above the 5σ chi-square band (a few per cent, converged to a local minimum) are dropped before variances are formed. “Variance” below is the total filtered Monte Carlo variance $\text{tr Cov}(\hat{\mathbf{w}})$.

run	script	light model	atom / η	OD	β
<code>decoherence_1</code>	<code>v5</code>	spin flips, separate readout	$\eta = 30$	$-(\sigma = 0.021)$	0–5
<code>onebeam_1</code>	<code>v6</code>	spin flips, one beam	$\eta = 30$	$\text{OD}_{\text{eff}} = 10$	0.5–20
<code>loss_1</code>	<code>v7</code>	loss + spin flips, one beam	Cs D2 $F = 3$	$\text{OD}_{\text{res}} = 140$	0.5–20
<code>loss_1_od10</code>	<code>v7</code>	same	same	$\text{OD}_{\text{res}} = 10$	0.5–20
<code>*_notensor</code>	<code>v7</code>	same beam, J_z^2 off	same	140 and 10	0.5–20

v5, separate readout. With $\beta = 0$ a decoherence-free closed system, strong nonlinearity wins only at short times: $\beta = 5$ reaches a variance of 10^{-2} at $\Omega t = 1.15$ against 1.52 for $\beta = 0$, and 10^{-3} at 1.86 against 2.23, but its bound saturates at $\sim 10^{-5}$ while the closed system keeps improving, so for $T \gtrsim 4/\Omega$ the best β is 0. This is the $\beta_{\text{opt}} \sim \eta/T$ picture, and it is the most pessimistic case for the nonlinear protocol because the reference carries no decoherence at all.

v6, one beam. With $\beta = 0$ excluded the optimum is finite at every T : by the bound the best β steps $20 \rightarrow 10 \rightarrow 5 \rightarrow 2 \rightarrow 1 \rightarrow 0.5$ at $\Omega t = 0.5, 0.9, 1.3, 2.7, 6.7$. This run also exposed the Cramér–Rao bookkeeping issue of Section 9: at $\beta \geq 5$ the Monte Carlo variance sat 20–50 times above $[\mathbb{E}F]^{-1}$ with no rejected fits and zero excess cost, which is not optimizer failure but the Jensen gap, $\mathbb{E}[F_i^{-1}]/\text{tr}[\mathbb{E}F]^{-1} = 2.4, 3.5, 17$ at $\beta = 2, 5, 20$.

v7, loss: the estimator reaches its bound. In `loss_1` the ratio of the Monte Carlo variance to $\text{tr} \mathbb{E}[F_i^{-1}]$ at T is 0.99–1.03 for every β ; the least-squares estimator with random control is efficient once its fits have converged (Fig. 1). The tensor-off references are efficient at weak probe but sit above their own reachable bound at strong probe ($17\times$ at $\beta = 20$ for $\text{OD}_{\text{res}} = 140$; $5\text{--}90\times$ for $\beta \geq 5$ at $\text{OD}_{\text{res}} = 10$), an identifiability problem of the purely linear dynamics that only makes the reference worse where the nonlinear run is already far ahead.

v7: the nonlinearity helps, and by a lot. The tensor-off runs share seeds, decoherence and shot noise with the tensor-on runs, so the comparison is paired. Table 1 gives, per run and target variance, the earliest time after which the variance stays below the target (minimized over β) and the β that achieves it. Every target is reached three to four times sooner with the nonlinearity, at both optical depths.

target tr Cov	10^{-2}	10^{-3}	10^{-4}	10^{-5}	10^{-6}
$\text{OD}_{\text{res}} = 140, J_z^2$ on	0.52 ($\beta=20$)	1.16 (20)	1.47 (20)	2.46 (10)	9.55 (2)
$\text{OD}_{\text{res}} = 140, J_z^2$ off	2.13 (10)	3.28 (10)	4.27 (10)	5.69 (2)	11.21 (2)
$\text{OD}_{\text{res}} = 10, J_z^2$ on	1.09 (20)	2.06 (10)	2.85 (10)	11.99 (2)	–
$\text{OD}_{\text{res}} = 10, J_z^2$ off	4.37 (5)	7.77 (2)	10.67 (1)	13.24 (2)	–

Table 1: Time (Ωt) after which the total filtered variance stays below the target, best over β (the optimal β in parentheses); – = not reached within $T = 14/\Omega$.

At equal β the ratio of variances with and without the J_z^2 term (Fig. 2) is 0.05, 0.0015 and 0.0013 at $\Omega t = 2$ for $\beta = 5, 10, 20$ ($\text{OD}_{\text{res}} = 140$), still 0.35 and 0.03 at $\Omega t = 8$ for $\beta = 10, 20$, and 0.03 at T for $\beta = 20$; for $\beta \leq 2$ it is 0.3–0.8 throughout. The ratio of the reachable bounds tells the same story (0.02–0.06 at $\Omega t = 2$ for $\beta \geq 5$, 0.5–0.8 at T) and is independent of OD, since σ^2 is a common factor. The naive bound $[\mathbb{E}F]^{-1}$ says the opposite, a ratio of 1.2–2.9 at T : the nonlinear dynamics make the information depend more strongly on the particular random control, which enlarges the Jensen gap and which a fixed-control bound cannot see. The mechanism is the one anticipated in Section 8: the nonlinearity converts the probe into information faster, so a target

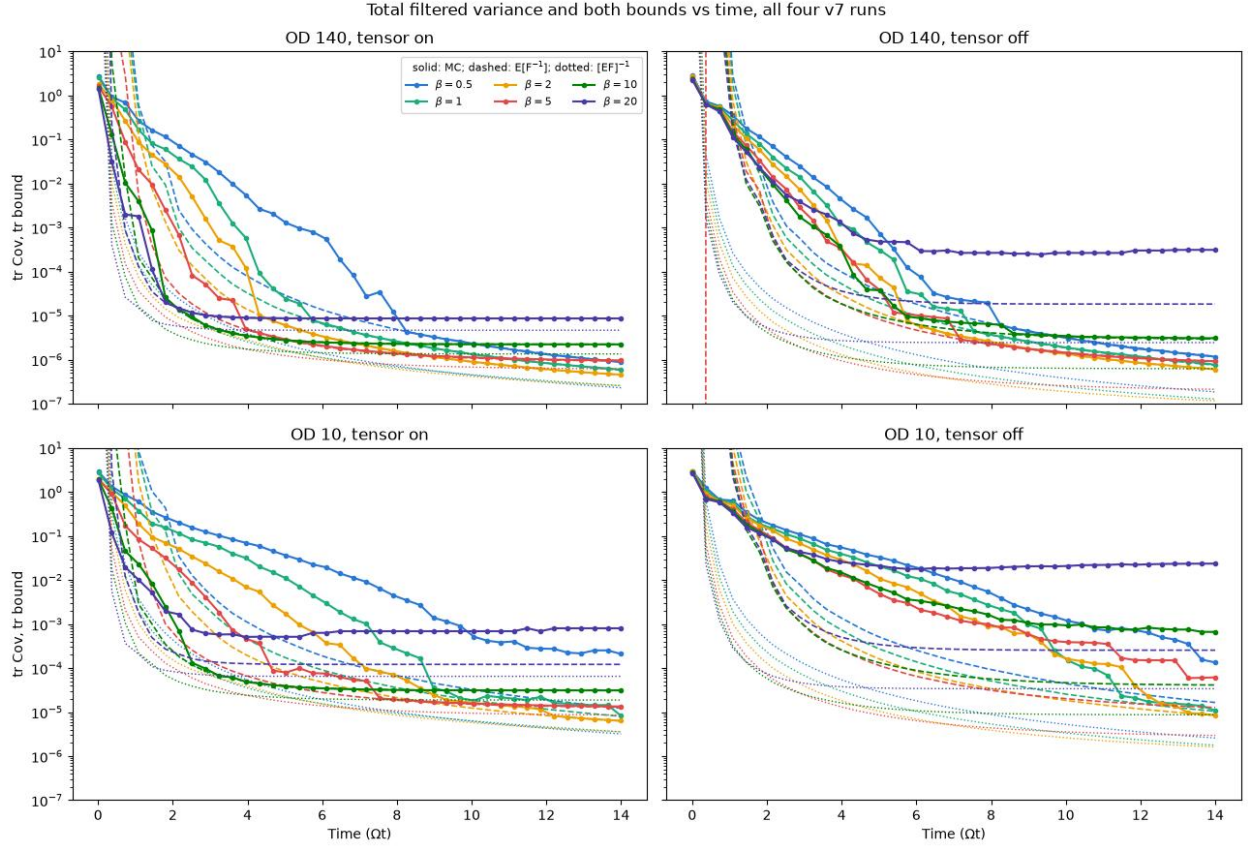


Figure 1: Total filtered Monte Carlo variance (solid), reachable bound $\text{tr} \mathbb{E}[F_i^{-1}]$ (dashed) and naive bound $\text{tr} [\mathbb{E}F]^{-1}$ (dotted) versus time for the four v7 runs. Left: with the J_z^2 term; right: identical beam without it. Top: $\text{OD}_{\text{res}} = 140$; bottom: 10.

precision is reached before the loss has eaten the signal. The best β by Monte Carlo falls with the time budget, 20 up to $\Omega t \approx 2$, 10 to ≈ 5 , 5 to ≈ 9 , then 2, in agreement with $\beta_{\text{opt}} \sim \eta_{\text{tot}}/T$ up to a prefactor.

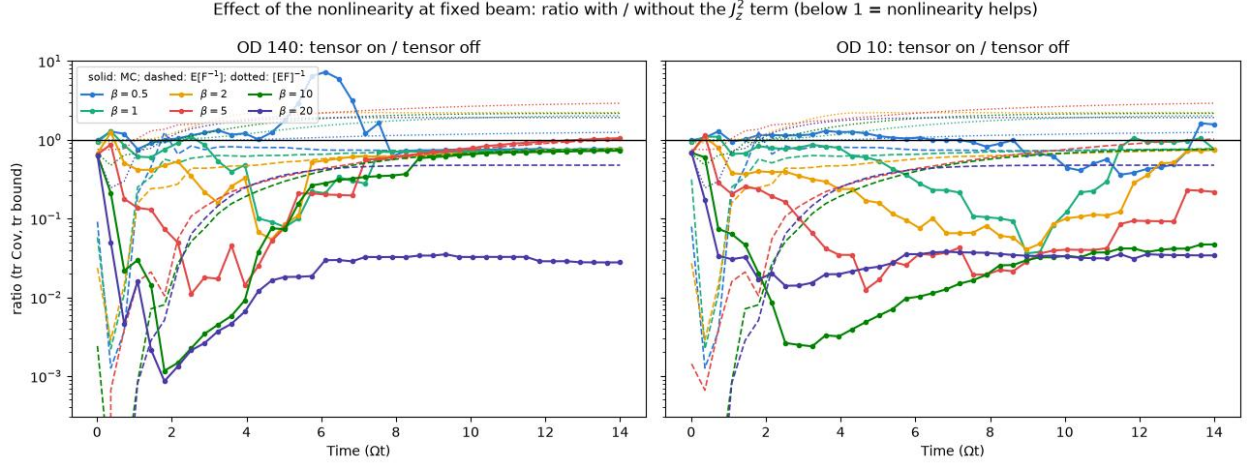


Figure 2: Effect of the nonlinearity at fixed beam: ratio of the run with the J_z^2 term to the run without it, versus time and per β . Solid: Monte Carlo variance; dashed: reachable bound; dotted: naive bound. Below 1 the nonlinearity helps.

v7: loss is the gentler decoherence. `loss_1` and `onebeam_1` have the same β list, the same nonlinearity per unit of initial signal relaxation ($\eta_{\text{tot}} \approx 30$) and shot noise equal to within 5%. They differ only in what the scattering does to the atoms. With loss-dominated relaxation the variance at T is lower by a factor 0.84, 0.44, 0.075 at $\beta = 0.5, 1, 2$ and by three orders of magnitude at $\beta \geq 5$ (Fig. 3). Survivors keep evolving coherently and only the signal amplitude decays, so the strong-probe regime that looked hopeless under the spin-flip model is usable. Since loss is 97 % of the real relaxation for Cs, the v5/v6 picture overstated the cost of the probe by a large factor.

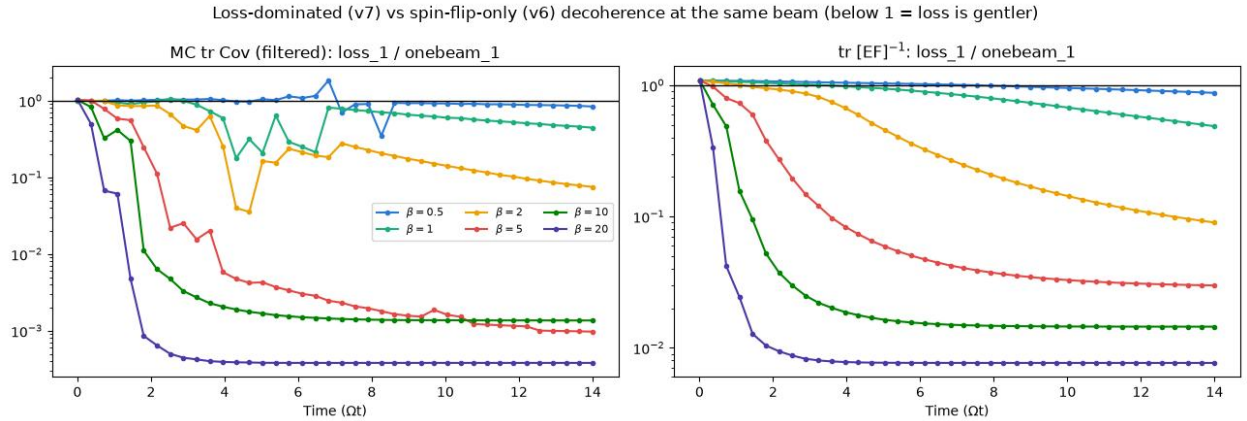


Figure 3: Loss-dominated (v7, `loss_1`) versus spin-flip-only (v6, `onebeam_1`) decoherence at the same beam: ratio of Monte Carlo variances (left) and of naive bounds (right).

v7: free-space optical depth. At $\text{OD}_{\text{res}} = 10$ the shot noise is 3.7 times larger and the variances at T about ten times larger (8×10^{-6} against 6×10^{-7} at $\beta = 1$); the tightest target is out of reach within the run, but the ordering of β and the gain from the nonlinearity are unchanged (Table 1).

What remains. (i) Atoms transferred to F' are treated as dark; they carry a vector response of opposite sign and a different light shift, and a faithful treatment needs the $F \oplus F'$ space. (ii) The tensor-off reference at strong probe does not reach its own bound; whether that is an optimizer or an identifiability limit of the linear dynamics is untested and does not affect the conclusions. (iii) Cs D1, $F = 4$ offers $\eta_{\text{tot}} = 408$, twelve times more nonlinearity per unit of relaxation than the D2, $F = 3$ case run here, at the cost of $J = 4$. (iv) The runs fix T and sweep β ; the experimental question is the converse, the shortest T for a required precision, which Table 1 answers only on the β grid.